THE PERSISTENT Π_1

Ximena Férnandez and Gonzalo Ortega

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& THE LOWER CENTRAL QUOTIENTS BARCODES

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THE LIMITATIONS OF H_1

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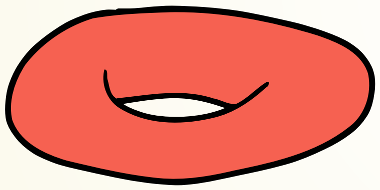

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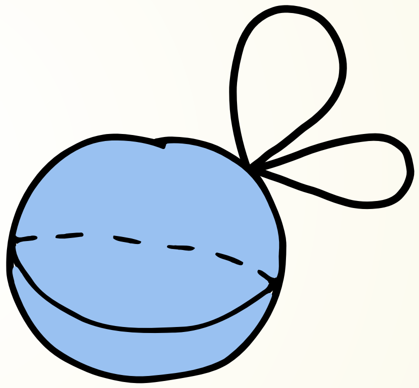
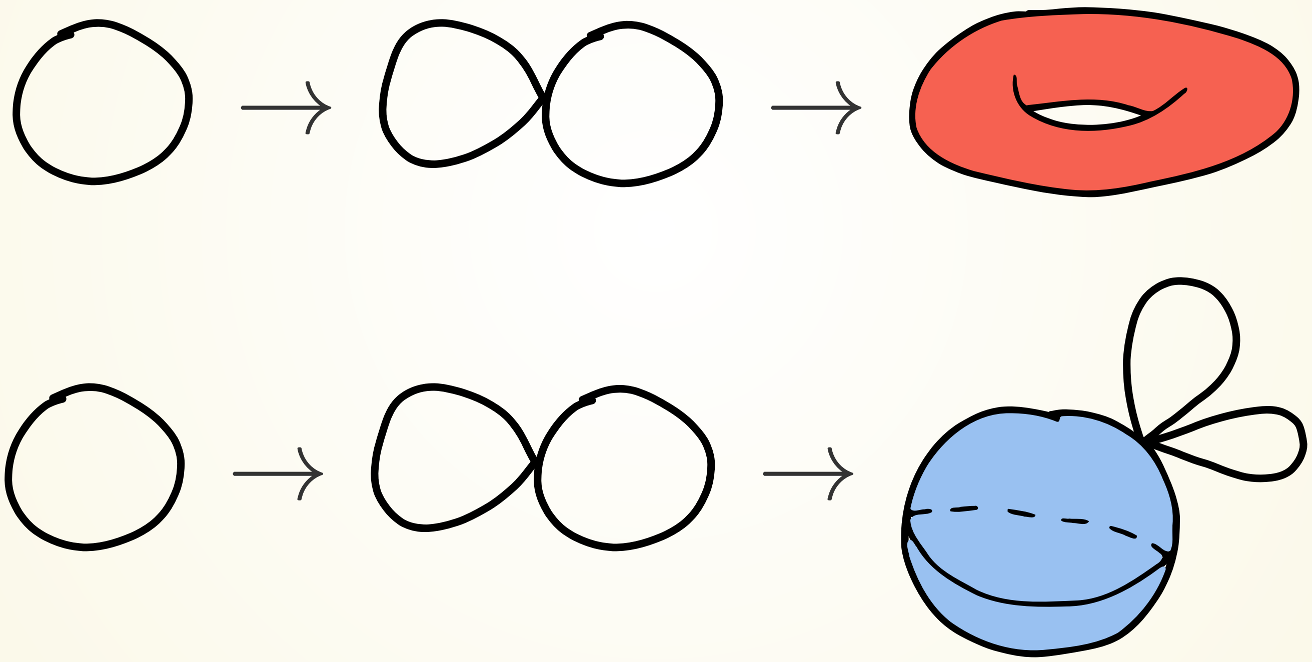


$S^2 \vee S^1 \vee S^1$

THE LIMITATIONS OF H_1



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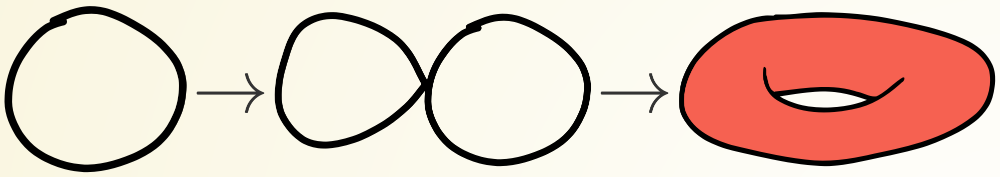
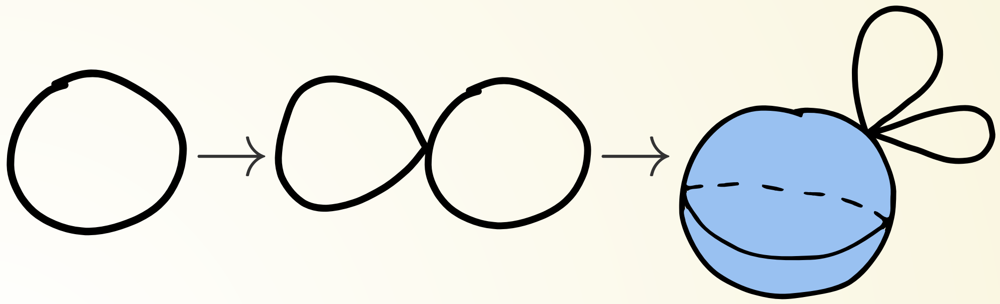
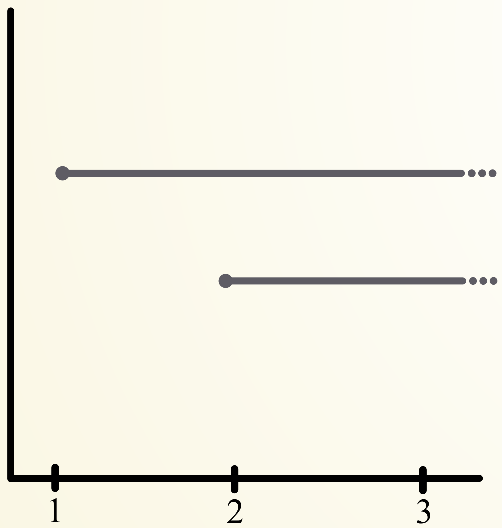
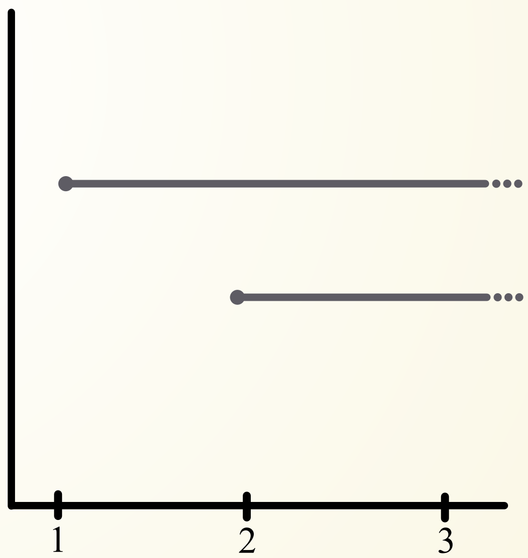




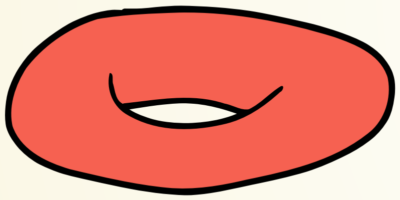


$H_1(T)$

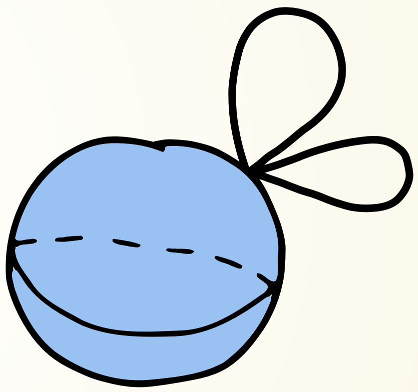


$H_1(S^2 \vee S^1 \vee S^1)$

THE LIMITATIONS OF H_1



$$\pi_1(T) = \mathbb{Z} \oplus \mathbb{Z}$$



$$\pi_1(S^2 \vee S^1 \vee S^1) = \mathbb{Z} * \mathbb{Z}$$

PERSISTENT Π_1

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Let K be a finite CW-complex, $f : K \to \mathbb{R}$ a filtration function. Let $x_0 \in K^0$. Let M be an acyclic matching in $\mathcal{X}(K)$.

Definition (Filtrated matching function):

$$\begin{aligned} \tilde{f} &: M \to \mathbb{R} \\ (e, e') &\mapsto \max\{f(e), f(e')\}. \end{aligned}$$

Denote

$$M_\alpha := \tilde{f}^{-1}((-\infty, \alpha)).$$

PERSISTENT Π_1

$$\begin{tikzcd}
(K_i, M_i) \arrow[r, "\phi_{i\to j}^{K,M}"] \arrow[d, "\pi_1"] & (K_j, M_j) \arrow[d, "\pi_1"] \\
\mathcal{P}_i \arrow[r, "\psi_{i\to j}"] & \mathcal{P}_j
\end{tikzcd}$$

PERSISTENT Π_1

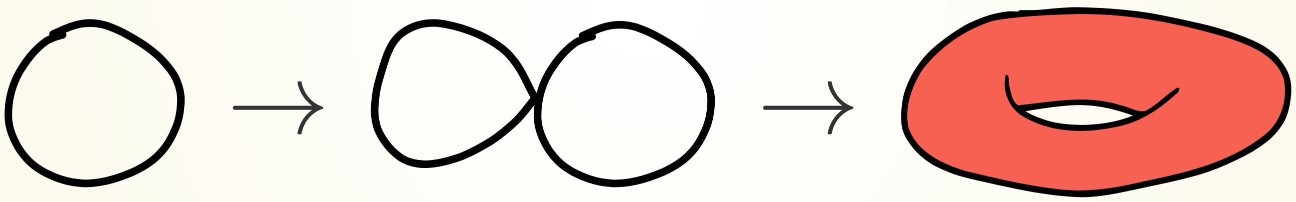
We define $\psi^S_{i\to j}\colon F(S_i) \to F(S_j)$ as

$$\psi^S_{i\to j}(e) = \begin{cases} 1 & \exists x \in K^0_j : (x,e) \in M^0_j \\ \psi^S_{i\to j}((w_1^{-1}w_2^{-1})^{\varepsilon}) & \exists \sigma = w_1 e^{\varepsilon} w_2 \in K^2_j, \\ & (e,\sigma) \in M^1_j,\ \epsilon \in \{1,-1\} \\ e & \text{else} \end{cases}$$

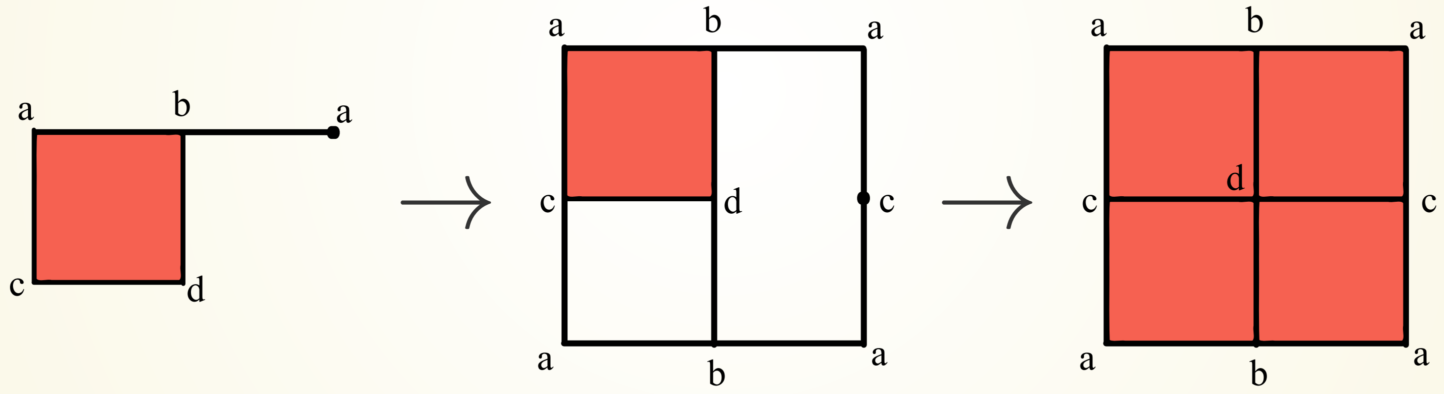
Were if $(w_1^{-1}w_2^{-1})^{\varepsilon} = e_1^{\varepsilon_1} \dots e_n^{\varepsilon_n}$,

$$\psi^S_{i\to j}((w_1^{-1}w_2^{-1})^{\varepsilon}) = \psi^S_{i\to j}(e_1^{\varepsilon_1}) \dots \psi^S_{i\to j}(e_n^{\varepsilon_n}).$$

Example: Persistent π_1 of a torus filtration



Example: Persistent π_1 of a torus filtration



LOWER CENTRAL QUOTIENTS

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Definition (Commutator): Let G be a group, $g, h \in G$,

$$[g, h] = g^{-1}h^{-1}gh.$$

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Definition (Lower central series):

$$G = \gamma_1(G) \trianglerighteq \gamma_2(G) \trianglerighteq \cdots \trianglerighteq \gamma_n(G) \trianglerighteq \ldots$$

where

$$\gamma_{n+1}(G) = [\gamma_n(G), G].$$

LOWER CENTRAL QUOTIENTS

Definition (Lower central quotient):

$$lcq_n(G) := \frac{\gamma_n(G)}{\gamma_{n+1}(G)}.$$

LOWER CENTRAL QUOTIENTS

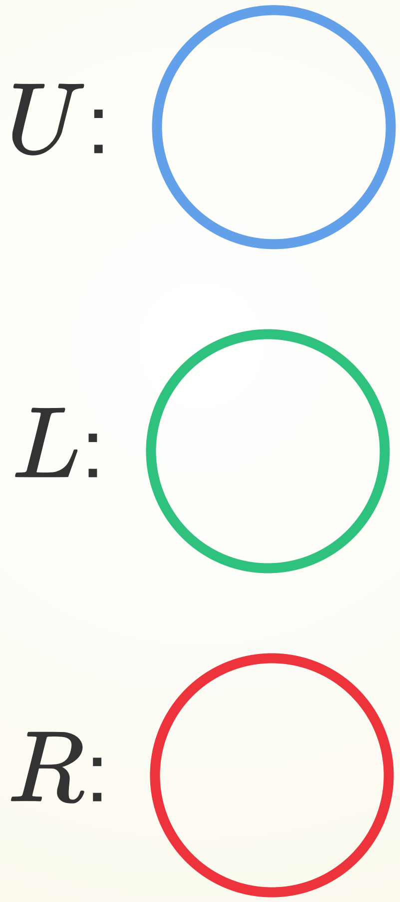
$$\begin{array}{ccc} \mathcal{P}_i & \xrightarrow{\psi_{i\to j}} & \mathcal{P}_j \\ \downarrow{lcq_n} & & \downarrow{lcq_n} \\ I_i & \xrightarrow{\xi_{i\to j}} & I_j \end{array}$$

LOWER CENTRAL QUOTIENTS

$$\begin{array}{ccc} \mathcal{P}_i & \xrightarrow{\psi_{i\to j}} & \mathcal{P}_j \\ \downarrow lcq_n & & \downarrow lcq_n \\ I_i & \xrightarrow{\xi_{i\to j}} & I_j \end{array}$$

Lemma: The map $\xi_{i\to j}: I_i \to I_j$ is a well-defined unique homomorphism.

Example: lcq_n barcodes of links



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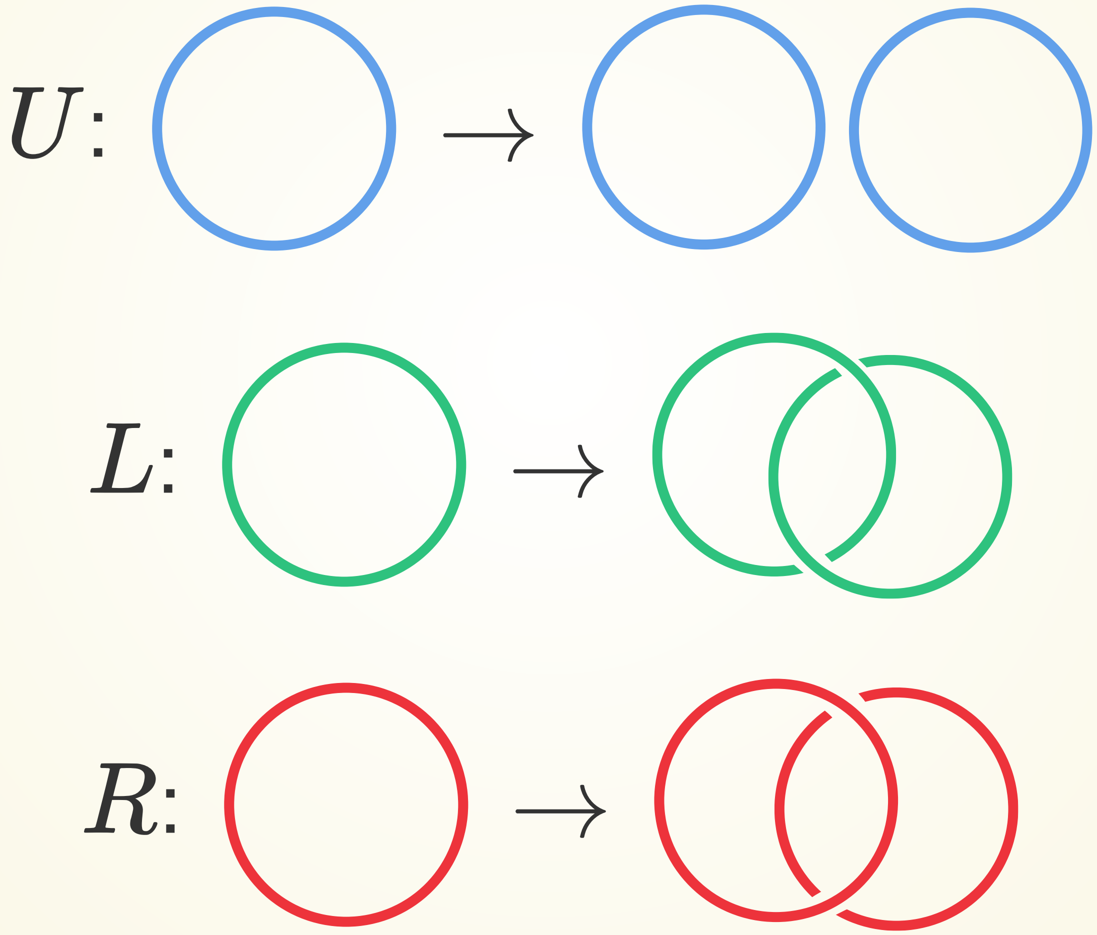

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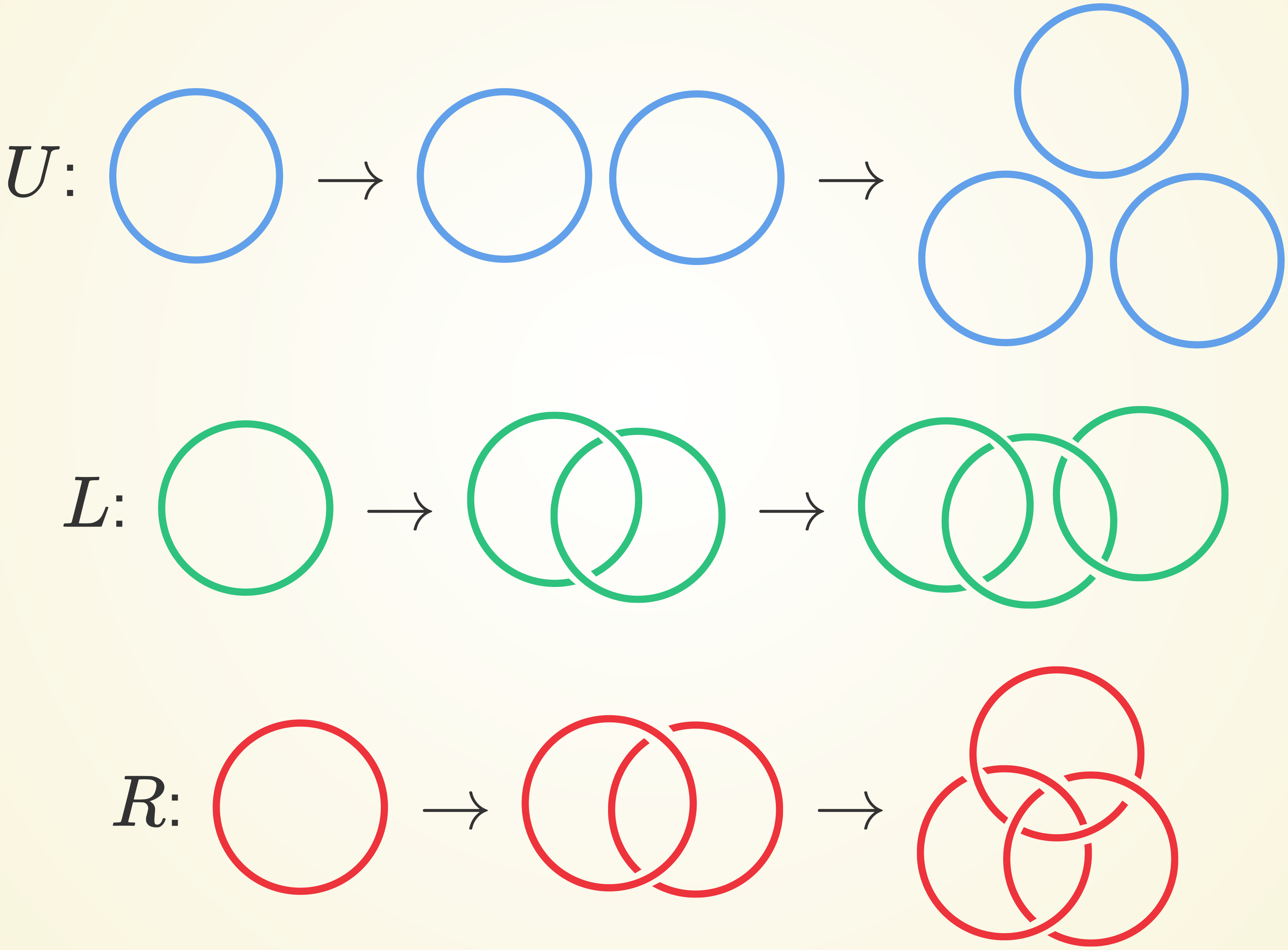
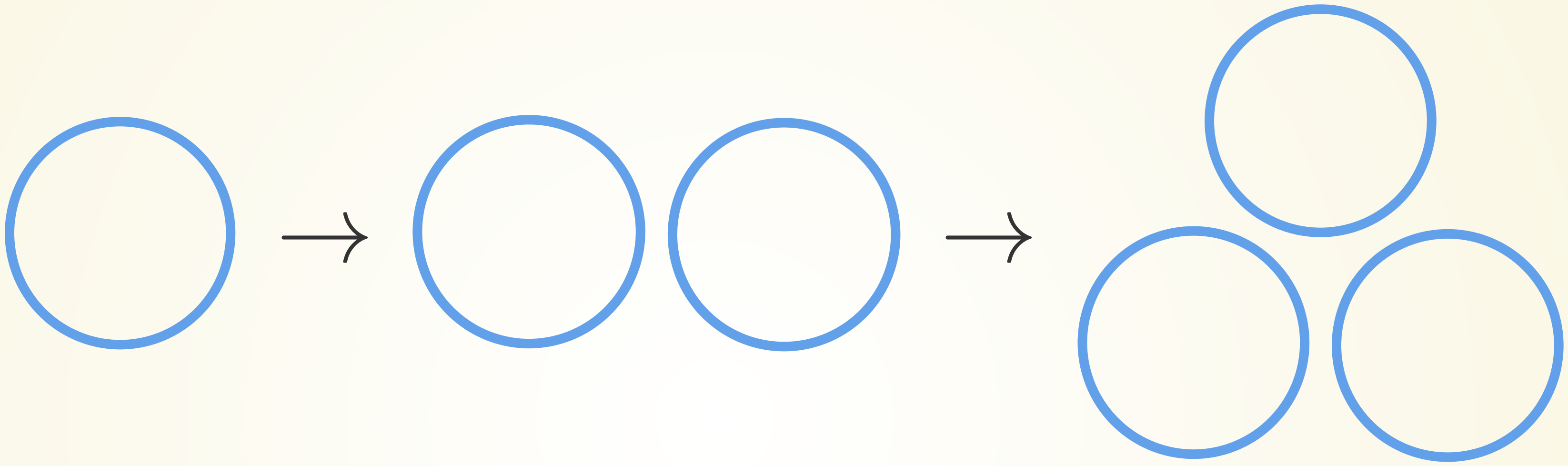


Example: lcq_n barcodes of links

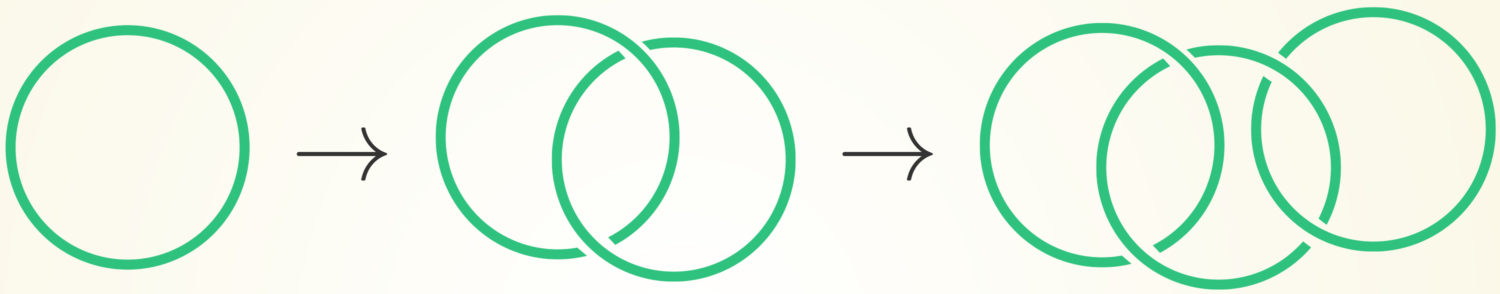
U:



$$\pi_1(\mathbb{R}^3 \setminus U_1) = \langle u_1 \mid - \rangle$$
$$\pi_1(\mathbb{R}^3 \setminus U_2) = \langle u_1, u_2 \mid - \rangle$$
$$\pi_1(\mathbb{R}^3 \setminus U_3) = \langle u_1, u_2, u_3 \mid - \rangle$$

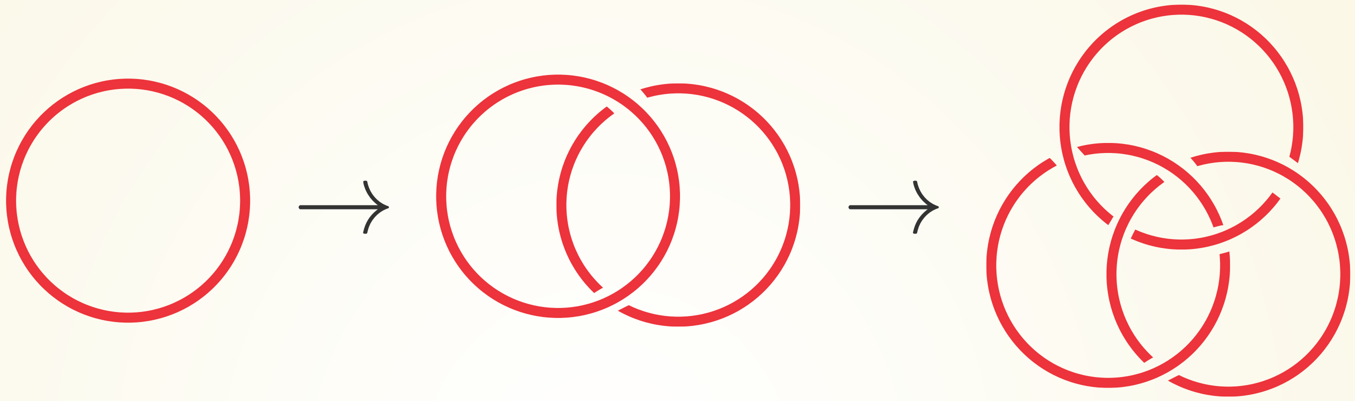
Example: lcq_n barcodes of links

L:



$$\pi_1(\mathbb{R}^3 \setminus L_1) = \langle l_1 \mid - \rangle$$
$$\pi_1(\mathbb{R}^3 \setminus L_2) = \langle l_1, l_2 \mid [l_1, l_2] \rangle$$
$$\pi_1(\mathbb{R}^3 \setminus L_3) = \langle l_1, l_2, l_3 \mid [l_1, l_2], [(l_1 l_3), l_2] \rangle$$

Example: lcq_n barcodes of links

R: 

$$\pi_1(\mathbb{R}^3 \setminus R_1) = \langle r_1 \mid - \rangle$$

$$\pi_1(\mathbb{R}^3 \setminus R_2) = \langle r_1, r_2 \mid - \rangle$$

$$\pi_1(\mathbb{R}^3 \setminus R_3) = \langle r_1, r_2, r_3 \mid r_3 r_2^{-1} r_3^{-1} r_1^{-1} r_3 r_2 r_3^{-1} r_2^{-1} r_1 r$$

$$r_3 r_2 r_3^{-1} r_1 r_3 r_1^{-1} r_2^{-1} r_1 r_3^{-1} r_1^{-1} \rangle$$

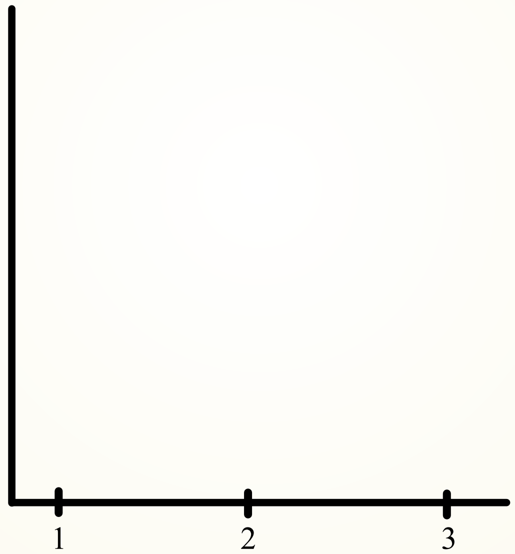
Example: lcq_n barcodes of links

	U_1	U_2	U_3
lcq_1	$\mathbb{Z}$	$\mathbb{Z}^2$	$\mathbb{Z}^3$
lcq_2	0	$\mathbb{Z}$	$\mathbb{Z}^3$
lcq_3	0	$\mathbb{Z}^2$	$\mathbb{Z}^8$

	L_1	L_2	L_3
lcq_1	$\mathbb{Z}$	$\mathbb{Z}^2$	$\mathbb{Z}^3$
lcq_2	0	0	$\mathbb{Z}$
lcq_3	0	0	$\mathbb{Z}^2$

	R_1	R_2	R_3
lcq_1	$\mathbb{Z}$	$\mathbb{Z}^2$	$\mathbb{Z}^3$
lcq_2	0	$\mathbb{Z}$	$\mathbb{Z}^3$
lcq_3	0	$\mathbb{Z}^2$	$\mathbb{Z}^6$

Example: lcq_n barcodes of links



Example: lcq_n barcodes of braids

Let P_n be the nth pure braid group.

$$P_4 \to P_3 \to P_2$$

GAP CODE

```
LoadPackage("nq");

LcsQuotient := function(group, i)
    local h, lcs;
    nq := NilpotentQuotient(group, i+1);
    lcs := LowerCentralSeries(nq);
    if i < Length(lcs) then
        return AbelianInvariants(lcs[i] / lcs[i+1]);
    fi;
    return TrivialGroup();
end;
```

REFERENCES